Answer Key: Transformers Fundamentals

Course: AI103

Part I: MCQ Answers

1. Answer: A

2. Answer: B

3. Answer: A

4. Answer: C

5. Answer: C

6. Answer: A

7. Answer: A

8. Answer: B

9. Answer: C

10. Answer: B

11. Answer: A

12. Answer: C

13. Answer: A

14. Answer: D

15. Answer: C

16. Answer: C

17. Answer: A

18. Answer: C

19. Answer: C

20. Answer: B

21. Answer: C

22. Answer: C

Part II: True / False Answers

1. Answer: False
Statement: Transformers are a type of recurrent neural network (RNN).

2. Answer: True
Statement: The transformer model uses self-attention mechanisms to draw global dependencies between input and output sequences.

3. Answer: False
Statement: The scaled dot-product attention mechanism is used to increase the dimensionality of the input.

4. Answer: True
Statement: Multi-head attention is used to allow the model to jointly attend to information from different representation subspaces.

5. Answer: False
Statement: The transformer model is only useful for natural language processing (NLP) tasks.

6. Answer: True
Statement: The attention function is computed as a weighted sum of the values.

7. Answer: True
Statement: The softmax function is used to assign weights to the values in the attention mechanism.

8. Answer: False
Statement: The transformer model uses convolutional layers to handle sequential data.

9. Answer: True
Statement: The decoder in the transformer model generates an output sequence of symbols one element at a time.

10. Answer: True
Statement: The transformer model is more complex than traditional RNNs.

Part III: Written Answers

1. What is the main difference between transformers and traditional RNNs?
Answer: The main difference between transformers and traditional RNNs is that transformers use self-attention mechanisms to draw global dependencies between input and output sequences, whereas RNNs use recurrent layers to process sequential data. This allows transformers to handle long-range dependencies more effectively and parallelize the computation more easily.

2. How does the self-attention mechanism work in transformers?
Answer: The self-attention mechanism in transformers works by computing the attention function as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. This allows the model to attend to different parts of the input sequence simultaneously and weigh their importance.

3. What is the purpose of the scaled dot-product attention mechanism in transformers?
Answer: The scaled dot-product attention mechanism is used to counteract the effect of large dot products, which can push the softmax function into regions with extremely small gradients. This is done by dividing the dot products by the square root of the dimensionality of the input.

4. How does multi-head attention work in transformers?
Answer: Multi-head attention in transformers works by linearly projecting the queries, keys, and values multiple times with different learned linear projections, and then performing attention on each of these projected versions in parallel. The outputs from each attention head are then concatenated and linearly transformed to produce the final output.

5. What are some common applications of transformers?
Answer: Transformers have been widely adopted for various applications, including language modeling, text classification, machine translation, and computer vision. They have also been used in other fields, such as audio processing and robotics, and have shown promising results.

6. How can transformers be used for text classification tasks?
Answer: Transformers can be used for text classification tasks by using the output of the transformer model as input to a classification layer. The transformer model can be trained to produce a continuous representation of the input text, which can then be used to predict the class label.

7. What are some common misconceptions about transformers?
Answer: Some common misconceptions about transformers include the idea that they are only useful for NLP tasks, that they are too complex to implement, and that they require large amounts of training data. However, transformers can be used for a wide range of tasks, can be implemented using pre-trained models and libraries, and can be trained on smaller datasets with careful tuning of hyperparameters.

8. What is the mathematical representation of the attention function in transformers?
Answer: The attention function in transformers is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. The mathematical representation of the attention function is $Attention(Q, K, V) = softmax(QK^T / \sqrt{dk}) * V$, where Q, K, and V are matrices representing the queries, keys, and values, respectively.

9. How does the transformer model handle sequential data?
Answer: The transformer model handles sequential data by using self-attention mechanisms to draw global dependencies between input and output sequences. This allows the model to attend to different parts of the input sequence simultaneously and weigh their importance, rather than processing the

sequence sequentially like traditional RNNs.

10. What is the advantage of using transformers for natural language processing tasks?
Answer: The advantage of using transformers for natural language processing tasks is that they can handle long-range dependencies more effectively and parallelize the computation more easily than traditional RNNs. This allows transformers to achieve state-of-the-art results on a wide range of NLP tasks, including language modeling, text classification, and machine translation.